



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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Computer Networking



❖ Network Topology

- Topology defines the structure or layout of a network, showing how computers/nodes and other network devices are interconnected or arranged with each other.
- Topology = Structure/Layout of a network.

There are two types of connections

1. point-to-point

2. multi-point

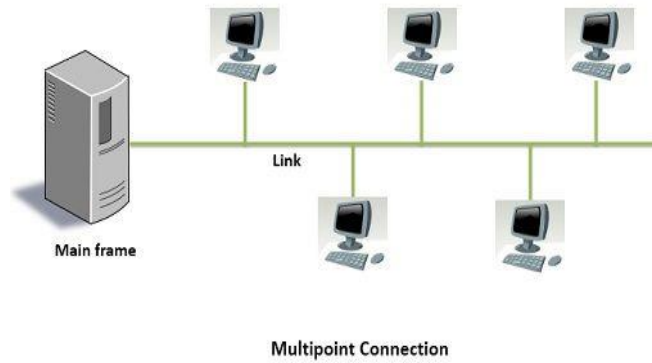
1. Point-to-point connections

- Point-to-point connection provides a dedicated and direct link between exactly two devices.
- for example : a computer connected directly to a printer, Router to Router

2. Multi-Point / multidrop connection

- Multipoint connection provides a single communication link that is shared by three or more devices on a network.

- └ Computer D



(Physical arrangement)

- ## 6. Hybrid Topology

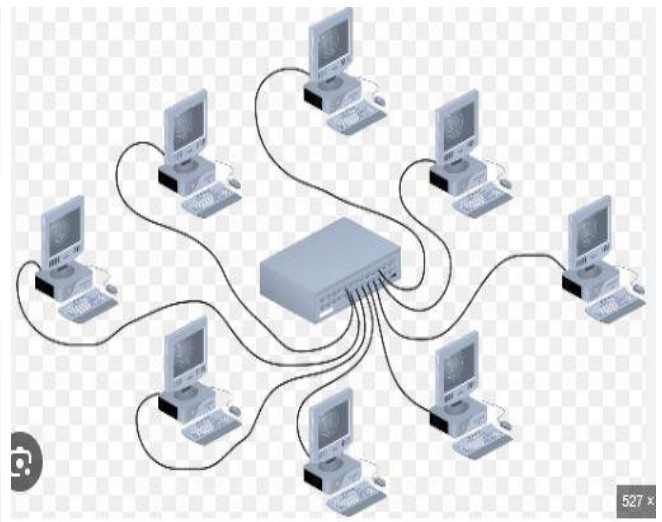
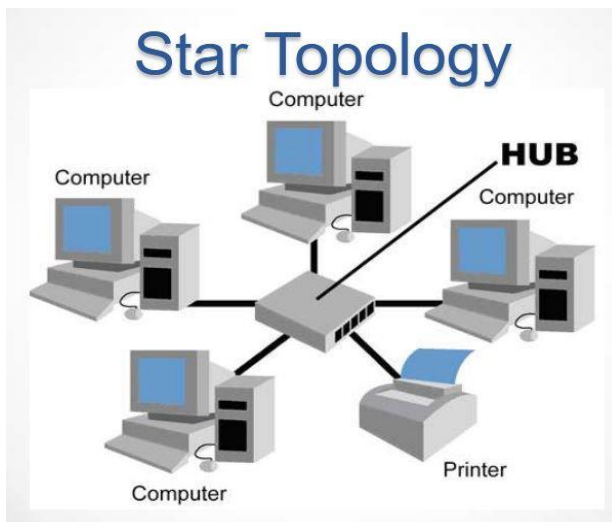


1. Star Topology

- The central device then forwards the data to Device 2.

Note :

Modern star networks generally use a **switch** rather than a hub. A hub broadcasts incoming data to all connected ports, while a switch forwards it to the intended destination port.



➤ Advantages

- Less expensive than mesh topology, since each device is connected only to the central hub/switch.
- Easy to install and configure.
- Requires less cabling than mesh topology.
- Reliable/Robust: If one link or cable fails, only that particular device is affected; other links remain active.
- Easy to identify and troubleshoot faults.
- Easy to add or remove devices without disrupting the entire network.

➤ Disadvantages

- Requires more cabling than Bus and Ring topologies, although it requires less cabling than Mesh topology.
- Dependency on the central device: The entire network depends on the central hub. If the hub fails, the entire network becomes unavailable/dead.

➤ Applications

- ✓ Star topology used in Local Area Networks(LANs).
- ✓ Home/office LAN, schools, computer lab

2. Mesh Topology

- Every device has a dedicated point-to-point link to every other device.
- Each node must be connected to $n - 1$ other nodes.
- A fully connected mesh with n devices requires $n(n - 1)/2$ physical

links/port (channels/wire).

- Each device must have $n - 1$ I/O ports to connect directly to all other devices.

Important Formulas

- Connections from one computer: $n - 1$
- Total physical links in a fully connected mesh: $n(n - 1)/2$
- Ports required per computer: $n - 1$

Q1. Total 6 computer(C1 to C6),C1 connect all computer (C2,C3,...,C6) how many use physical wire /Ports require.

Ans. – Total computer $n= 6$, total connected $6-1=5$

Q2. Total 10 computer,how many use physical connection/wire all computer fully connected each other ?

Ans. – total computer $n=10$

Fully connected physical wire required= $10(10-1)/2,...$

Answer – 45 wires

Q3. There are 8 computers (C1–C8). C1 is connected to all other computers. How many physical links/ports are required?

Ans : 7 links/ports

Q4. There are 12 computers. All computers are fully connected to each other. How many physical wires are required?

Ans :66 wires

Q5. There are 15 computers. Each computer is directly connected to every other computer. How many ports are required for each computer?

Ans : 14

Q6. A fully connected mesh network has 28 physical links. How many computers are there?

Ans : 8

Q7. fully connected mesh network has 78 physical links. How many computers are connected in the network?

Ans : 13

Q8. A fully connected mesh network has 15 computers. If 3 computers are removed, how many physical links are removed?

Ans : 39

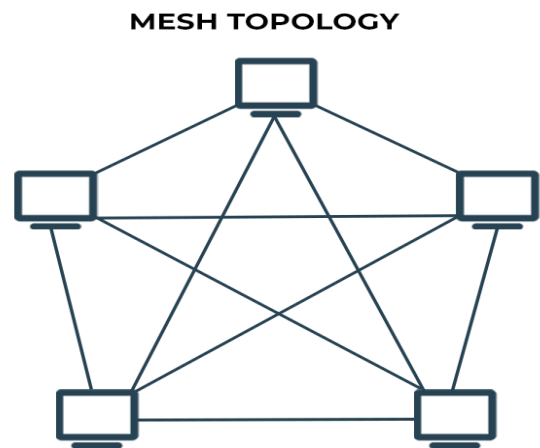
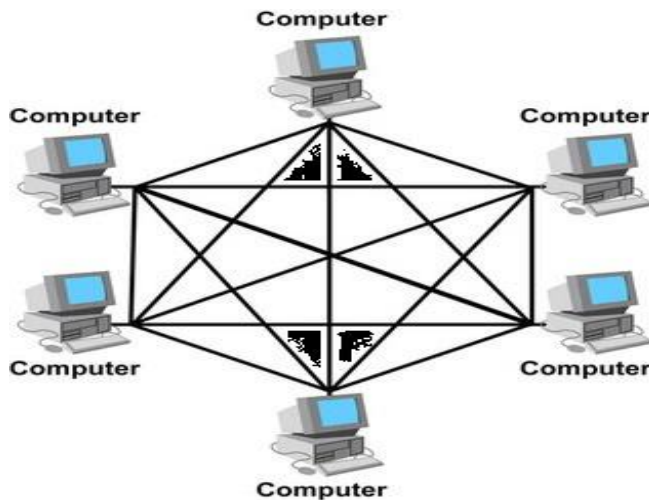
Q9. A fully connected mesh network requires 190 physical links. If 5

additional computers are added, how many new physical links will be required?

Ans : 110

Q10. A fully connected mesh network has 16 computers. Each computer is connected to every other computer. If each connection uses 3 meters of cable, what is the total cable required?

Ans : 360 meters



➤ Advantages

- They use dedicated links, so each link can only carry its own data load. Thus, traffic problems can be avoided.
- It is robust/strong. If any one link gets damaged, it does not affect the other links.
- It provides privacy and security, as the message travels along a dedicated link.
- Fault identification and fault isolation are easy.

➤ Disadvantages

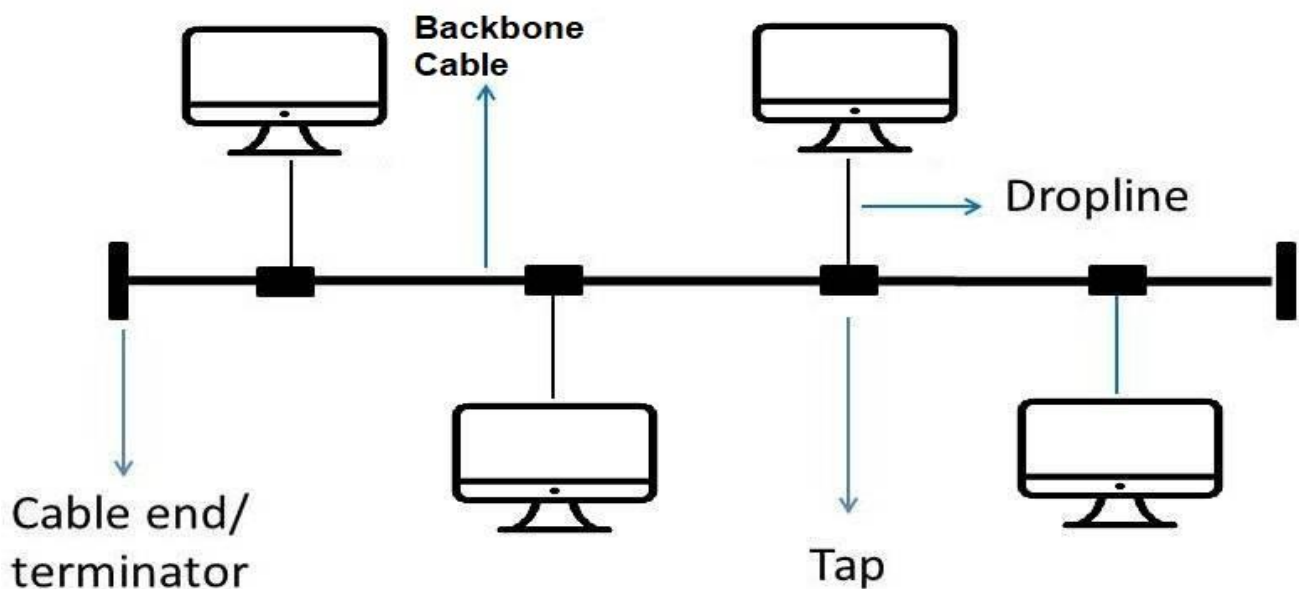
- The amount of cabling and the number of I/O ports required are very large, since every device is connected to every other device through dedicated links.
- The large amount of wiring may exceed the available space.
- The hardware required to connect each device is highly expensive.

➤ Applications

- ✓ Telephone Regional office (Customer care).
- ✓ WAN(Wide Area Network).
- ✓ Backbone networks, wireless mesh topology

3. Bus Topology

- A bus topology is a multipoint connection.
- Here, one long cable acts as a backbone to which all devices are connected through **drop lines** and **taps**.
- **Drop line**: A connection between a device and the backbone cable.
- **Tap**: A connector used to connect a device/drop line to the backbone cable. In traditional bus networks, T-connectors (T-taps) and BNC connectors were commonly used.
- Only one device can transmit data at a time on the shared bus.
- When a node wants to send a message over the network, it broadcasts the message onto the bus. All nodes connected to the bus receive the signal, whether or not the message is addressed to them.



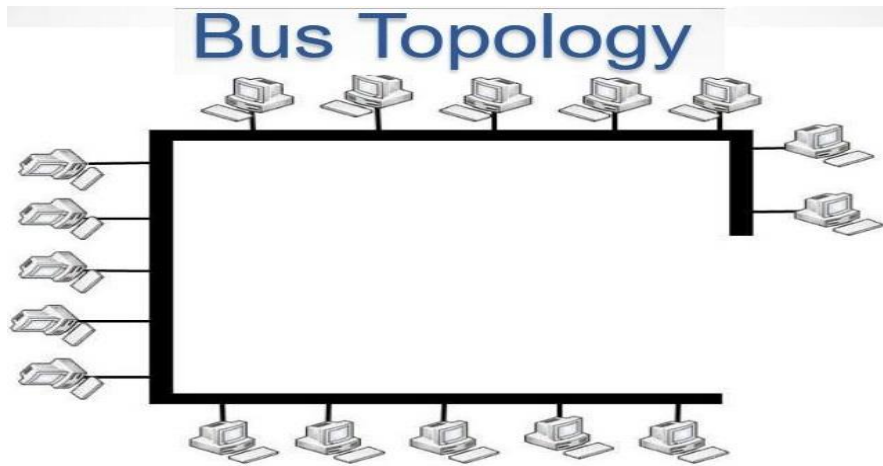
➤ Advantages

- Ease of installation
- Less cabling

➤ Disadvantages

- Difficult to reconfigure and isolate faults.
- Difficult to add new devices to the network.
- Attenuation is the loss or weakening of a signal during transmission. Repeaters are used to regenerate the signal.
- If there is any fault in the backbone cable, it can stop all transmissions on the network.
- If two nodes send messages simultaneously, their signals collide

with each other.

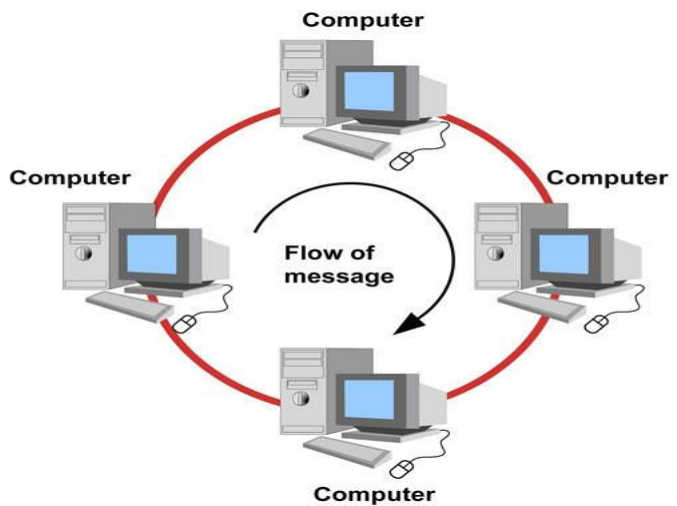
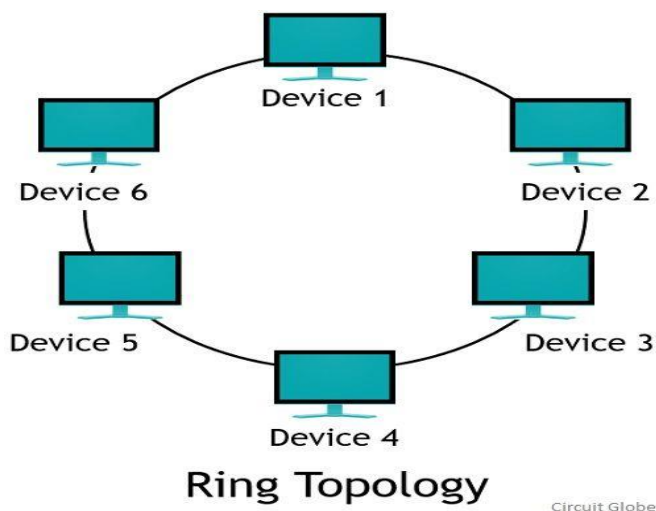


➤ Applications

- ✓ Small temporary network, older ethernet networks.

4. Ring Topology

- Ring topology is like a bus topology, but with connected ends.
- Connection type : Point to point connection
- The node that receives the message from the previous computer will node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.
- The most common access method of the ring topology is –
- Token passing: It is a network access method in which token is passed from one node to another node. So, it works on **token passing technology**.
- Token: It is a a frame that circulates around the network.
- It is also called the **circular topology**.



➤ **Advantages:**

- Easy to install.
- Easy to reconfigure.
- Fault identification is easy.

➤ **Disadvantages:**

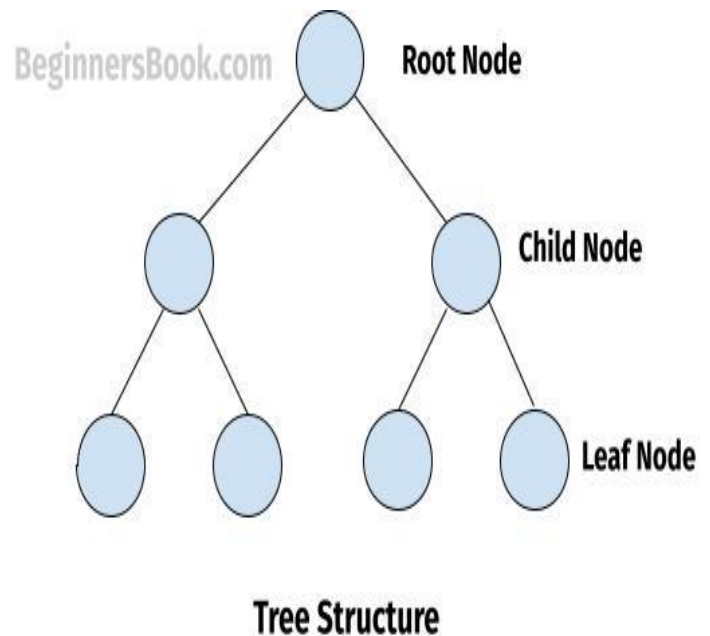
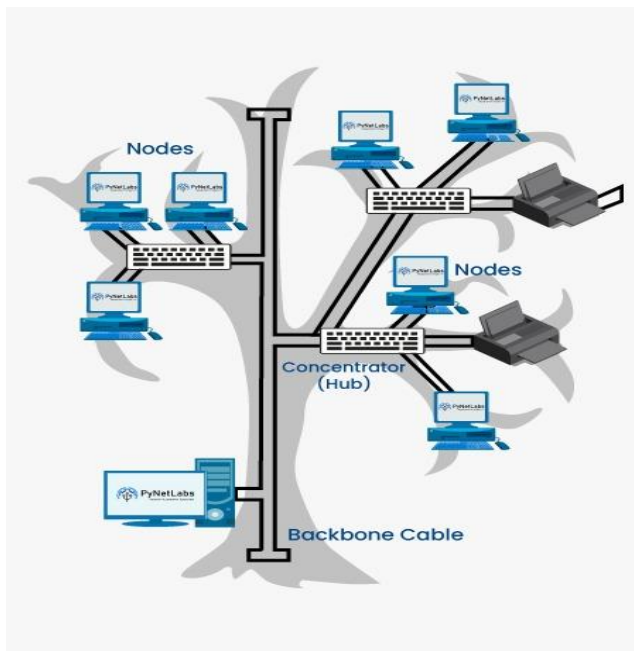
- Unidirectional traffic.
- Break in a single ring can break entire network.
- To add or delete a device requires effect two connections.

➤ **Applications:**

- ✓ Ring topologies are found in some office buildings or school campuses.
- ✓ Today high speed LANs made this topology less popular.
- ✓ Some industrial networks.

5. Tree Topology

- Tree topology combines the characteristics of bus topology and star topology
- Connection type : Point to point connection
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion
- The top-most node in tree topology is known as a root node, and all other nodes are the Child of the root node
- There is only one path exists between two nodes for the data transmission. Thus, it forms a parent-child hierarchy.



➤ **Advantage**

- It has point-to-point wiring for individual divided/segments.
- We can easily add the new device to the existing network.
- In tree topology, the whole network is divided into segments known as star networks which can be easily managed and maintained
- Error detection and error correction are very easy in a tree topology

➤ **Disadvantage**

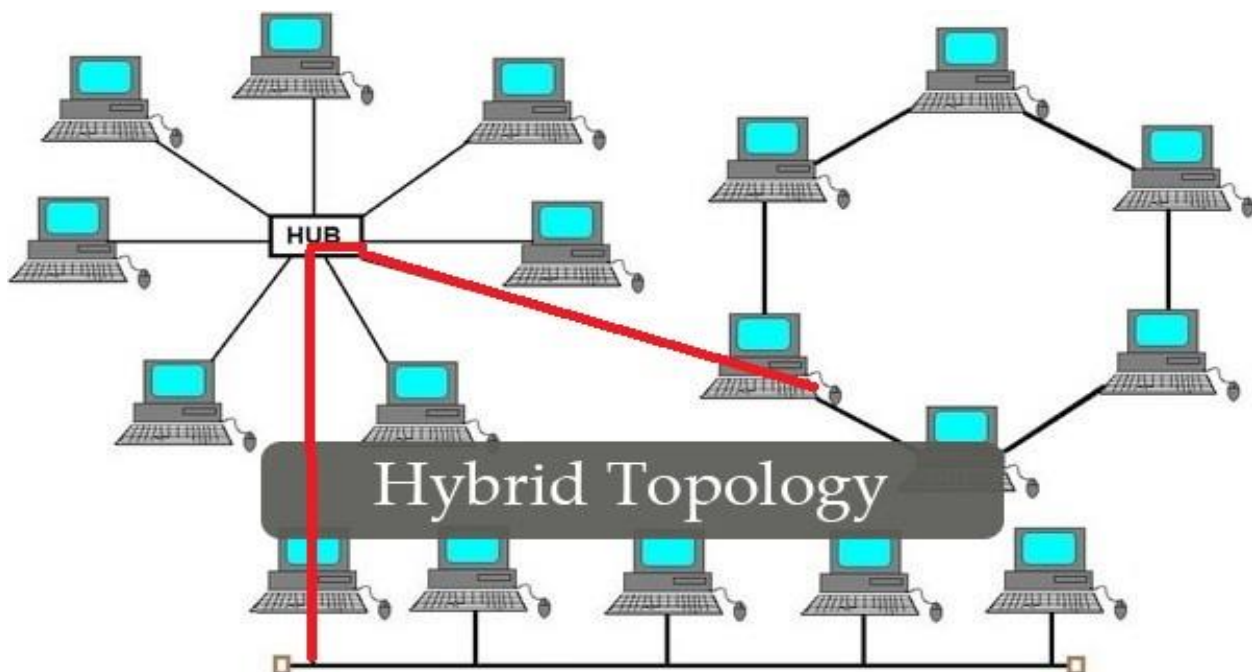
- A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.
- Root Node damage all child nodes network loss.
- Devices required for broadband transmission are very costly.

➤ **Application**

- ✓ Large organizations, campus networks

6. Hybrid Topology

- A network which contain two or more topology of physical structure and connected under a single backbone channel, is called hybrid topology.
- Connection type : Depends on the topology used.



➤ Advantage

- New topologies and nodes can be added and removed from the network easily.
- If there occurs any error in the network then it is detected easily and that network device or node can be exchanged with a new device or node.
- The characteristics of each topology are combined in the hybrid network and weakness of different topologies are eliminated(दूर करना/हटाना).

➤ Disadvantage

- Designing hybrid networks is a complex process. Hybrid networks are difficult in installing and configuring.
- If the backbone of the network is damaged then network performance is affected also.
- A hybrid topology is used in making large networks so it needs more cables and cooling system.

➤ Application of hybrid topology

- ✓ Large enterprise networks
- ✓ Data centers
- ✓ Complex organizational networks.

❖ Two types of topology based on physical arrange and data flow

1. Physical topology

2. Logical topology

1. Physical topology

- The complete physical structure of the cable (or transmission media) is called the physical topology.
- The physical topology of a network refers to the configuration of cables, computers, and other peripherals.
- Example of Physical topology : Star, mesh, ring, bus, tree, hybride

2. Logical topology

- The way data flows through the network (or transmission media) is called the logical topology. A logical topology is the method used to pass information between workstations.
- Example : Bus, Ring

Key points

Question	Best Choice
Very Cheapest	Bus
Very reliable / Fault tolerant	Mesh
Modern LAN practical	Star
Easy installation	Star
Easy troubleshooting	Star
High privacy/security	Mesh
Less cabling	Bus
Easy expansion	Star / Tree
Large hierarchical network	Tree
Most flexible	Hybrid
Most expensive	Mesh
Collision possibility	Bus
Central device dependency	Star
Backbone failure affects network	Bus

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